G01	Basic HTML	Default CSS	Attributes	Definition
01	DOCTYPE	None	None	# defines the document type # all HTML documents must start with a (DOCTYPE) declaration # the declaration is not an HTML tag # it is an information to the browser about what document type to expect # the (DOCTYPE) declaration is NOT case-sensitive # in HTML5, the declaration is simple - <! DOCTYPE html > # in older (HTML4 or XHTML), the declaration is more complicated - because the declaration must refer to a DTD (document type definition) # HTML 4.01 - <! DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" - "http://www.w3.org/TR/html14/loose.dtd" > # XHTML 1.1 - <! DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" - "http://www.w3.org/TR/xhtml111/DTD/xhtml111.dtd"
02	html	Display: block html: focus - Outline: none	Xmlns Global	# defines an HTML document and is used once in a document # the (html) tag represents the root of an HTML document # the (html) tag is the container of all HTML element (except DOCTYPE) # NB: should always include the (lang) attribute inside the (html) tag - to declare the language of the web page to assist -search and browser
03	head	Display: none	Global	# contains metadata/information for the document and used once in a document # the (head) element is a container for metadata (data about data) - and is placed between the (html) tag and the (body) tag # metadata is data about the HTML document. Metadata is not displayed # metadata typically define (title, character set, styles, scripts, meta info) # the following elements can go inside the (head) element - title | style | base | link | meta | script | noscript
04	title	Display: none	Global	# defines a title for the document and is required and used once only # the title must be text-only, and it is shown in the (title bar) # it is important for SEO - defines a tile in the browser toolbar - provides a title for the page when it is added to favorites - displays a title for the page in search-engine results # good titles - descriptive title (avoid one or two word titles) - search engines will display about (50-60) characters of the title - make the title as accurate and meaningful as possible

05	body	Display: block Margin: 8px Body: focus – Outline: none	Global | Event	# defines the document's body and is used once in a document # the (body) element contains all the contents of an HTML document
06	h1 – h6	Display: block Font-size: 2em Margin-top: 0.67em Margin-bottom: 0.67em Margin-left: 0 Margin-right: 0 Font-weight: bold	Global | Event	# Defines HTML headings # headings are titles or subtitles that you want to display on a webpage # headings are defined with the (h1 to h6) tags # (h1) defines most important heading and (h6) defines least important # search engines use it to index the structure and content of web page # users often skim a page by its headings, is important for doc structure # NB: headings for headings only not to use for (BIG or bold) text # NB: each heading has a default size but can specify the size
07	p	Display: block Margin-top: 1em Margin-bottom: 1em Margin-left: 0 Margin-right: 0	Global | Event	# Defines a paragraph
08	br	None	Global | Event	# Inserts a single line break, not to add space between paragraphs # is useful for writing addresses or poems # is an empty tag which means that it has no end tag
09	hr	Display: block Margin-top: 0.5em Margin-bottom: 0.5em Margin-left: auto Margin-right: auto Border-style: inset Border-width: 1px	Global | Event	# Defines a thematic break or change in the content # is displayed as a horizontal rule that is used to separate content
10	<!-- -->	None	None	# Defines a comment and not displayed in the browsers # is used to insert comments in the source code # use comments to explain code that can help to edit source code later - This is especially useful if you have a lot of code # can use the comment to (hide) scripts from browsers # NB: (//) at the end of comment line is the JS comment symbol - this prevents JS from executing the (-->) tag

G02	Meta Info	Default CSS	Attributes	Definition
01	meta	None	Charset | Content | http-equiv | name Global	# defines metadata about an HTML document # metadata is data (information) about data # (meta) tags always go inside the (head) element, and used to specify - character set | page description | keywords | document author | viewport # metadata will not be displayed on the page, but is machine parsable # metadata is used by browsers (how to display content or reload page) - search engines (keywords), and other web services # there is a method to let web designers take control over the viewport # the viewport is the user's visible area of a web page # it varies with the device – it will be smaller on a mobile phone than pc # should include the following (meta) element in all your web pages - < meta name="viewport" content="width=device-width, initial-scale=1.0" > - this gives the instructions on how to control page's dimensions | scaling - the (width=device-width) part sets the width of the page of follow the device - the (initial-scale-1.0) part sets the initial zoom level when page is first loaded
02	base	None	Href | Target Global | Event	# specifies the base URL/target for all relative URLs in a document # (base) tag must have either an (href) or a (target) attribute present, or both # only be one single (base) element in a document, must inside (head)
03	basefont (css)	None	None	# specifies a default color, size, and font for all text in a document # not in HTML5 | use CSS

G03	Semantics, and Layout	Default CSS	Attributes	Definition
01	div	Display: block	Global | Event	# defines a section in a document # it defines a division or a section in an HTML document # it is used as a container for HTML elements - which is then styled with CSS or manipulated with JS # the (div) tag is easily styled by using the (class | id) attribute # any sort of content can be put inside the (div) tag
02	span	None	Global | Event	# defines a section in a document # the (span) tag is an inline container used to mark up of a text - or a part of a document # it is easily styled by CSS or manipulated with JS using the (class | id) attribute # the (span) tag is much like the (div) element, - but (div) is a (block-level) element and (span) is an (inline)
03	header	Display: block	Global | Event	# defines a header for a document or section # the (header) element represents a container for introductory - content or a set of navigational links # a (header) element typically contains - one or more (h1 – h6) | logo or icon | authorship information # NB: you can have several (header) elements in one document # NB: it cannot be placed within (footer | address | header)
04	footer	Display: block	Global | Event	# defines a footer for a document or section # a (footer) element typically contains - authorship information | copyright information - contact info | sitemap | back to top links | related documents - NB: contact info inside a (footer) should go inside (address) # you can have several (footer) elements in one document
05	main		Global | Event	# specifies the main content of a document # the content inside the (main) element should be unique to doc # it should not contain any content that is repeated across doc - such as (sidebars | navigation | copyright info | logo | search) # NB: there must not be more than one (main) element in doc - the (main) element must NOT be a descendant of an - (article | aside | footer | header | nav) element
06	section	Display: block	Global | Event	# defines a section in a document
07	search	Display: block	Global | Event	# defines a search section # here comes a set of elements that is related to search # elements inside (search) can typically be (form) used to search # it does not render as anything special, but can use CSS

08	article	Display: block	Global | Event	# defines an article (independent | self-contained) content # an article should make sense on its own - it should be possible to distribute it independently from rest # potential sources for the (article) element - forum post | blog post | news story # NB: (article) does not render as anything special in browser - can use CSS to style it
09	aside	Display: block	Global | Event	# defines content aside from the pate content # it defines some content aside from the content it is placed in # aside content should be indirectly related to surrounding content # NB: the (aside) content is often placed as a sidebar in a doc # NB: (aside) element does not render as anything special in doc - can use CSS to style (aside) element
10	details	Display: block	Open Global | Event	# defines additional details that user can view or hide on demand # the (details) tag is often used to create an interactive widget - that the user can open and close, by default, the widget is closed - when open, it expands, and displays the content within # any sort of content can be put inside the (details) tag # NB: the (summary) tag is used in conjunction with (details) to - specify a visible heading for the details
11	dialog	None	Open Global | Event	# defines a dialog-box or subwindow # (dialog) makes it easy to create popup dialogs and modals
12	summary	Display: block	Global | Event	# defines a visible heading for a (details) element - the heading can be clicked to view/hide the details #NB: (summary) should be the first child element of (details)
13	style	Display: none	Media | Type Global | Event	# defines style information (CSS) for a document # inside (style) element, specify how elements render in browser # NB: if some properties have been defined for the same - selector in different style sheets, value from the last will use # to link to an external style sheet, use the (link) tag
14	data	None	Value Global	# adds a machine-readable translation of a given content # it provides both a machine-readable value for data processors, - and a human-readable value for rendering in browser # if content is (time- | date-related) use the (time) element instead

G04	Text Formatting	Default CSS	Attributes	Definition
01	b	Font-weight: bold	Global | Event	# defines bold text without any extra importance # NB: according to the HTML5 specification, - (b) tag should be used as a LAST resort - when no other tag is more appropriate. - headings should be denoted with the (h1 – h6) | emphasized (em) - important (strong) | marked/highlighted (mark)
02	strong	Font-weight: bold	Global | Event	# defines text with strong importance, content inside is typically (blold)
03	small	Font-size: smaller	Global | Event	# Defines smaller text (like copyright and other side-comments) # NB: this is not deprecated, but for richer effect use CSS
04	i	Font-style: italic	Global | Event	# defines a part of text in an alternate voice or mood. # The content inside is typically displayed in (italic) # it is often used to indicate a - technical term, a phrase from another language, a thought, etc # use (i) only when there is not a more appropriate semantic element - em | strong | mark | city | dfn
05	em	Font-style: italic	Global | Event	# defines emphasized text, the content inside is typically displayed in italic # a screen reader will pronounce the words in (em) without an emphasis
06	mark	Background-color: yellow Color: black	Global | Event	# defines marked/highlighted text
07	del	Text-decoration: line-through	Cite | Datetime Global | Event	# defines text that has been deleted from a document - browsers will usually strike a line through deleted text # NB: also look at the (ins) tag to markup instead text
08	ins	Text-decoration: underline	Cite | Datatime Global | Event	# defines a text that has been instead into a document - Browsers will usually underline instead text # NB: also look at the (del) tag to markup deleted text
09	sub	Vertical-align: sub Font-size: smaller	Global | Event	# defines subscripted text # subscript text appears half a character below the normal line, - and is sometimes rendered in a smaller font # use the (sub) to define superscripted text
10	sup	Vertical-align: super Font-size: smaller	Global | Event	# defines superscripted text # superscript text appears half a character above the normal line, - and is sometimes rendered in a smaller font # use the (sub) tag to define subscript text
	Text Quotations	Default CSS	Attributes	Definition
11	abbr	Display: inline	Global | Event	# defines an abbreviation or an acronym, (HTML | CSS | Dr. | ASAP | ATM) # use the global (title) attribute to show the description for - the abbreviation/acronym when mouse over the element

12	address	Display: block Font-style: italic	Global | Event	# defines contact info for the author/owner of a document/article # contact info can be an (email | URL | address | phone | social media) # the text in the (address) element usually renders in (italic)
13	blockquote	Display: block Margin-top: 1em Margin-bottom: 1em Margin-left: 40px Margin-right: 40px	Cite Global | Event	# defines a section that is quoted from another source # browsers usually indent (blockquote) element
14	q	Display: inline Q:before – Content: open-quote Q:after – Content: close-quote	Cite Global | Event	# the (q) tag defines a short quotation # browsers normally insert quotation marks around the quotation
15	cite	Font-style: italic	Global | Event	# defines the title of a work # (cite) tag defines the title of a creative work (poem | song | movie | painting) # NB: a person's name is not the title of a work # the text in the (cite) element usually renders in (italic)
16	bdo	Unicode-bidi: bidi-override	Dir Global | Event	# the bdo tag is used to overrides the current text direction # BDO stands for Bi-Directional Override
	Text Code	Default CSS	Attributes	Definition
17	code	Font-family: monospace	Global | Event	# defines a piece of computer code # the content inside is displayed in the browser's default monospace font # NB: is not deprecated, it is possible to achieve richer effect by using CSS
18	kbd	Font-family: monospace	Global | Event	# defines keyboard input # the content inside is displayed in the browser's default monospace font # NB: is not deprecated, it is possible to achieve richer effect by using CSS
19	samp	Font-family: monospace	Global | Event	# defines sample output from a computer program # the content inside is displayed in the browser's default monospace font # NB: is not deprecated, it is possible to achieve richer effect by using CSS
20	var	Font-style: italic	Global | Event	# defines a variable in programming or in a mathematical expression # the content inside is typically displayed in italic # NB: is not deprecated, it is possible to achieve richer effect by using CSS
21	pre	Display: block Font-family: monospace White-space: pre Margin: 1em 0	Global | Event	# defines preformatted text # text in a (pre) is displayed in a fixed-width, - and preserves both spaces and line breaks # the text will be displayed exactly as written in the HTML source code
	Others	Default CSS	Attributes	Definition
22	bdi	None	Global | Event	# isolates a part of text that might be formatted in different # BDI stands for Bi-Directional Isolation

				# (Bdi) tag isolates a part of text that might be formatted in a different from # useful when embedding user-generated content with unknown text direction
23	dfn	Font-style: italic	Global | Event	# specifies a term that is going to be defined within the content # (dfn) stands for the (definition element), and it specifies a term that is going - to be defined within the content # the nearest parent of (dfn) must also contain the definition for the term # the term inside the (dfn) tag can be any of the following – - just as the content of the (dfn) element) - with the title attribute added - with an (abbr) tag inside the (dfn) element - with the (id) attribute added. Then, it can refer the definition with an (a) tag
24	meter	None	Form | High | Low Max | Min | Value Optimum Global | Event	# defines a scalar measurement within a known range or a fractional value - this is also known as a gauge # NB: (meter) tag should not be used to indicate progress (in a progress bar) # NB: always add the (label) tag for best accessibility practices
25	progress	None	Max | Value Global | Event	# the (progress) tag represents the completion progress of a task # NB: always add the (label) tag for best accessibility practices # NB: use (progress) tag in conjunction with JS to display the progress of a task # NB: the (progress) tag is not suitable for representing a gauge - to represent a gauge, use the (meter) tag instead
26	ruby	Line-height: normal	Global | Event	# defines a ruby annotation (for East Asian typography) # a ruby annotation is a small extra text, attached to the main text to indicate - the pronunciation or meaning of the corresponding characters # this kind of annotation is often used in (Japanese) publications # use (ruby) together with (rt | rp) – same as below
27	rt	Line-height: normal	Global | Event	# defines an explanation/pronunciation of characters (East Asian typography) # the (rt) tag defines an explanation or pronunciation of characters in a ruby - for East Asian typography # use (rt) together with (ruby | rp) – - an (rt) element that gives that information, and an optional (rp) element that - defines what to show for browsers that not support ruby annotations
27	rp	None	Global | Event	# defines what to show in browsers that do not support ruby annotations # the (rp) tag can be used to provide parentheses around a ruby text, - to be shown by browsers that do no support ruby annotations # use (rp) together with (ruby | rt) – - (ruby) consists of one or more characters that needs an - explanation/pronunciation - an (rt) element that gives that information, and an optional (rp) element that - defines what to show for browsers that not support ruby annotations

29	s	Text-decoration: line-through	Global | Event	# defines text that is no longer correct # the (s) tag specifies text that is no longer correct, accurate or relevant # the text will be displayed with a line through it # the (s) tag should not be used to define deleted text in a document, use (del)
30	template	None	Global	# defines a container for content that should be hidden when the page loads # (template) tag is used as a container to hold some HTML content hidden from - the user when the page loads # the content inside (template) can be rendered later with JS # use (template) if have some HTML code you want to use over and over again, - But not until you ask for it. To do this without (template) tag, have to create - the HTML code with JS to prevent the browser from rendering the code
31	time	None	Datetime Global | Event	# the (time) tag defines a specific time (or datetime) # the (datetime) attribute of this element is used translate the time into a - machine-readable format so that browsers can offer to add date reminders - through the user's calendar, - and search-engines can produce smarter search results
32	u	Text-decoration: underline	Global | Event	# defines text that is unarticulated and styled differently from normal text # such as misspelled words or proper names in Chinese text # the content inside is typically displayed with an underline # can change this with CSS # avoid using the (u) element where it could be confused for a hyperlink
33	wbr	None	Global | Event	# defines a possible line-break # the (wbr) tag specifies where in a text it would be ok to add a line-break # NB: when a word is too long, the browser might break it at the wrong place - can use the (wbr) element to add word break opportunities
34	acronym (abbr)	None	None	# defines an acronym - not in HTML5 | use (abbr) # an acronym or abbreviation should be marked up with the (abbr) tag
35	big (css)	None	None	# defines big text - not in HTML5 | use CSS # specify different font-size for HTML elements (with CSS)
36	center (css)	None	None	# defines centered text - not in HTML5 | use CSS # center-align text (with CSS)
37	font (css)	None	None	# defines font, color, and size for text - not in HTML5 | use CSS # set the color of text (with CSS)
38	strike (del/s)	None	None	# defines strikethrough text - not in HTML5 | use (del | s) # use the (del) tag to define deleted text
39	tt (css)	None	None	# defines teletype text - not in HTML5 | use CSS # consider the (kbd | var | code | samp) element, or use CSS instead

G05	Lists	Default CSS	Attributes	Definition
01	menu	Display: block List-style-type: disc Margin-block-start: 1em Margin-block-end: 1em Margin-inline-start: 0px Margin-inline-end: 0px Padding-inline-start: 40px	Global | Event	# defines an alternative unordered list of content # use the (menu) tag together with the (li) tag to create menu items # NB: (menu) tag is an alternative to the (ul) tag - and browsers will treat these two lists equally
02	ul	Display: block List-style-type: disc Margin-top: 1em Margin-bottom: 1em Margin-left: 0 Margin-right: 0 Padding-left: 40px	Global | Event	# the (ul) tag defines an unordered list # use the (ul) tag together with the (li) tag to create unordered lists
03	ol	Display: block List-style-type: decimal Margin-top: 1em Margin-bottom: 1em Margin-left: 0 Margin-right: 0 Padding-left: 40px	Reversed Start | type Global | Event	# the (ol) tag defines an ordered list which can be numerical or alphabetical # the (li) tag is used to define each list item
04	li	Display: list-item	Value Global | Event	# the (li) tag defines a list item # the (li) tag is used inside (ol) list, (ul) list and (menu) list # in (ul) and (menu), the list items will usually be displayed with bullet points # in (ol), the list items will usually be displayed with numbers or letters
05	dir (ul)	None	None	# the (dir) tag defines a directory list - not in HTML5 | use (ul)
06	dl	Display: block Margin-top: 1em Margin-bottom: 1em Margin-left: 0 Margin-right: 0	Global | Event	# the (dl) tag defines a description list # the (dl) tag is used in conjunction with (dt) and (dd)
07	dt	Display: block	Global | Event	# the (dt) tag defines a term/name in a description list # the (dt) tag is used in conjunction with (dl) and (dd)
08	dd	Display: block Margin-left: 40px	Global | Event	# the (dd) tag defines a description of a term/name in a description list # the (dd) tag is used in conjunction with (dl) and (dt) # inside a (dd) tag can put paragraphs, line breaks, images, links, lists, etc

G06	Tables	Default CSS	Attributes	Definition
01	table	Display: table Border-collapse: separate Border-spacing: 2px Border-color: gray	Global | Event	# the (table) tag defines an HTML table # an HTML table consists of one (table) element and one or more (tr | th | td) element # (tr) defines a table row, (th) defines a table header, and (td) defines a table cell # an HTML table may also include (caption, colgroup, thead, tfoot, tbody) elements
02	caption	Display: table-caption Text-align: center	Global | Event	# the (caption) tag defines a table caption # the (caption) tag must be inserted immediately after the (table) tag # NB: by default, a table caption will be center-aligned above a table - however, the CSS properties (text-align) and (caption-side) can be used to align - and place the caption
03	tr	Display: table-row Vertical-align: inherit Border-color: inherit	Global | Event	# the (tr) tag defines a row in an HTML table # a (tr) element contains one or more (th | td) elements
04	th	Display: table-cell Vertical-align: inherit Font-weight: bold Text-align: center	Abbr | colspan Headers | rowspan Scope Global | Event	# the (th) tag defines a header cell in an HTML table # an HTML table has two kinds of cells - header cells – contains header information (created with the (th) element) - data cells – contains data (created with the (td) element) # the text in (th) elements are bold and centered by default # the text in (td) elements are regular and left-aligned by default
05	td	Display: table-cell Vertical-align: inherit	Colspan | headers Rowspan Global | Event	# the (td) tag defines a cell in an HTML table # an HTML table has two kinds of cells – same as above
06	thead	Display: table-header-group Vertical-align: middle Border-color: inherit	Global | Event	# the (thead) tag is used to groups the header content in a table # (thead) is used in conjunction with (tbody | tfoot) to specify each part of a table - (header | body | footer) # use these to enable scrolling of table body independently of header and footer # also, when printing a large table that spans multiple pages, these elements can - enable the table header and footer to be printed at the top and bottom of each page # NB: the (thead) element must have one or more (tr) tags inside # the (thead) tag must be used in the following context: - as a child of a (table), after any (caption) and (colgroup), - and before any (tbody | tfooter | tr) element # NB: (thead | tbody | tfoot) will not affect the layout of the table by default, use CSS
07	tbody			# the (tbody) tag is used to groups the body content in a table # the (tbody) is used in conjunction with (thead | tfooter) # the (thead) tag must be used in the following context: - as a child of a (table), after any (caption) and (colgroup), and (thead) elements # same as above

08	tfoot	Display: table-footer-group Vertical-align: middle Border-color: inherit	Global | Event	# the (tfooter) is used to groups the footer content in a table # the (tfooter) element is used in conjunction with the (thead | tbody) # the (tfooter) tag must be used in the following context – - as a child of a (table) element - after any (caption | colgroup | thead | tbody) elements
09	col	Display: table-column	Span Global | Event	# the (col) tag specifies column properties for each column within a (colgroup) # the (col) tag is useful for applying styles to entire columns, instead of repeating - the styles for each cell, for each row
10	colgroup	Display: table-column-group	Span Global | Event	# the (colgroup) tag specifies a group of one or more columns in a table for formatting # (colgroup) is useful for applying styles to entire columns, - instead of repeating the styles for each cell, for each row # the (colgroup) tag must be a child of a (table), after any (caption) - and before any (thead | tbody | tfoot | tr) elements # NB: to define different properties to a column within a (colgroup) use (col) within it

G07	Links and File Paths	Default CSS	Attributes	Definition
01	a	A:link, a:visited – Color: (internal value) Text-decoration: underline Cursor: auto A:link:active, a:visited:active – Color: (internal value)	Download | href Hreflang | media Ping | referrerpolicy Rel | target | type Global | Event	# (a) defines a hyperlink, which is used to link from one page to another # most important attribute of (a) is (href), which indicates destination # by default, links will appear as follows in all browsers - an unvisited link is underlined and blue - a visited link is underlined and purple - an active link is underlined and red # NB: if (a) has no (href) attribute, it is only a placeholder for hyperlink - a linked page is normally displayed in the current browser window, - unless you specify another target
02	link	Display: none	Crossorigin | href Hreflang | media Referrerpolicy | rel Sizes | title | type Global | Event	# (link) defines relationship between document and external resource # it is most often used to link to external style-sheets or to add favicon # it is an empty element, it contains attributes only
03	nav	Display: block	Global | Event	# the (nav) tag defines a set of navigation links # NB: notice that NOT all links of a document should be inside (nav) - the (nav) element is intended only for major blocks of navigation links # browsers, such as screen readers for disabled users, - use this to determine whether to omit initial rendering of this content

G08	Images	Default CSS	Attributes	Definition
01	img	Display: inline-block	Alt | crossorigin Height | ismap |loading longdesc | Width Referrerpolicy | size Src | srcset | usemap Global | Event	# the (img) tag is used to embed an image in an HTML page # images are not technically inserted into a web page, these are linked to web page - the (img) tag creates a holding space for the referenced image # the (img) tag has two required attributes - src – specifies the path to the image - alt – specifies an alternate text for the image, if width and height are not specified, - the page might flicker while the image loads # NB: to link an image to another document, simply nest (img) tag inside (a) tag
02	map	Display: inline	Name Global | Event	# the (map) tag is used to defines a client-side image map - an image map is an image with clickable areas # required (name) attribute of (map) is associated with - (img) usemap attribute and creates a relationship between image and map # (map) contains a number of (area) tag, that defines clickable areas in image map
03	area	Display: none	Alt | cords | download Href | hreflang | Media referrerpolicy | Rel shape | target | Type Global | Event	# (area) tag defines an area inside an image map - An image map is an image with clickable areas - (area) elements are always nested inside a (map) tag # NB: (usemap) attribute in (img) is associated with (map) element's - (name) attribute, and creates a relationship between image and map
04	canvas	Height: 150px Width: 300px	Height | width Global | Event	# (canvas) tag is used to draw graphics, on the fly, via scripting (usually JS) # (canvas) tag is transparent, and is only a container for graphics - you must use a script to actually draw the graphics # any text inside (canvas) element will be displayed in browsers with JS - disabled and in browsers that do not support (canvas) # NB: learn more about (canvas) element from HTML Canvas
05	figcaption	Display: block	Global | Event	# (figcaption) tag defines a caption for a (figure) element # (figcaption) element can be placed as the first or last child of (figure)
06	figure	Display: block Margin-top: 1em Margin-bottom: 1em Margin-left: 40px Margin-right: 40px	Global | Event	# the (figure) tag specifies self-contained content, like – - illustrations, diagrams, photos, code listings, etc # while the content of the (figure) is related to the main flow, - its position is independent of the main flow, - and if removed it should not affect the flow of the document # NB: (figcaption) is used to add a caption for (figure) element
07	picture	None	Global | Event	# defines a container for multiple image resources # (picture) gives web developers more flexibility in specifying image resources # most common use of (picture) will be for art direction in responsive designs - instead of having one image that is scaled up or down based on viewport width, - multiple images can be designed to more nicely fill the browser viewport

				# the (picture) contains two tags: one or more (source) and one (img) tag # browser will look for first (source) where media query matches viewport width, - and then it will display the proper image (specified in the (srcset) attribute) - (img) is required as the last child of (picture), - as a fallback option if none of source tags matches # NB: (picture) works similar to (video | audio) - set up different sources, and first source that fits preferences is one being used
08	svg	None	None	# the (svg) tag defines a container for SVG graphics # SVG has several methods for drawing (paths, boxes, circles, text and graphics)

G09	Audio | Video	Default CSS	Attributes	Definition
01	audio	None	Autoplay | controls | Loop muted | Preload | src Global | Event	# defines sound content # the (audio) tag is used to embed sound content in a document, - Such as music or other audio streams # (audio) contains one or more (source) tags with different audio sources - browser will choose the first source it supports # text between (audio) tags will only be displayed in browsers - that do not support the (audio) element # there are three supported audio formats in HTML (MP3 | WAV | OGG)
02	source	None	Media | sizes | Src Srcset | Type Global | Event	# defines multiple media resources for media elements (video | audio | picture) # the (source) tag allows you to specify alternative (video | audio | image) files - which the browser may choose from, based on browser support or viewport width - the browser will choose the first (source) it supports
03	track	None	Default | kind | Label Src | Srclang Global | Event	# defines text tracks for media element (video | audio) # it is used to specify subtitles, caption files or other files containing text, - That should be visible when the media is playing - tracks are formatted in WebVTT format (.vtt files)
04	video	None	Autoplay | controls | width Height | Loop | Muted Poster | Preload | srec Global | Event	# the (video) tag is used to embed video content in a document, - such as (movie clip or other video streams) # (video) contains one or more (source) tags with different video source - browser will choose the first source it supports # text between (video) tag will not be displayed in browsers and not support (video) tag # there are three supported video formats in HTML (MP4 | WebM | OGG)

G10	Forms	Default CSS	Attributes	Definition
01	form	Display: block Margin-top: 0em	Accept-charset | Action Autocomplete | Enctype Method | Name novalidate | Rel | target Global | Event	# defines an HTML form for user input # (form) tag is used to create an HTML form for user input # (form) can contain one or more of the following form elements – - input | textarea | button | select | option | optgroup - fieldset | label | output
02	input	None	Accept | alt | Autocomplete autofocus | Checked | dirname Disabled | form | Formaction formenctype | Formmethod formnovalidate | Formtarget height | list | Max | maxlength Min | Minlenght | Multiple name | Pattern | Placeholder Readonly | required | Size src | step | type | Value width | Popovertarget Popovertargetaction Global | Event	# defines an input control # (input) tag specifies an input filed where the user can enter data # (input) is the most important (form) element # (input) can be displayed in several ways, depending on (type) # the different input types are as follows – - button | checkbox | color | date | datetime-local | email | file - hidden | image | month | number | password | radio | range - reset | search | submit | tel | text | time | url | week # NB: always use the (label) tag to define labels for these attributes - text | checkbox | radio | file | password
03	textarea	None	Autofocus | cols | dirname Disabled | form | Maxlenght Name | Placeholder Readonly | Required rows | wrap Global | Event	# the (textarea) tag defines a multi-line text input control # it is often used in a form, to collect user like comments or reviews # (textarea) can hold unlimited number of characters with fixed-width font # size of a text are is specified by the (cols | rows) attributes or CSS # (name) attribute is needed to reference the form data after - the form is submitted (if omit (name), no data will be submitted) # the (id) attribute is needed to associate the text area with a label # always add the (label) tag for best accessibility practices
04	button	None	Autofocus | disabled Form Formaction | Formenctype formmethod | Formtarget Formnovalidate | value Popovertarget | Name Type | Popovertargetaction Global | Event	# the (button) tag defines a clickable button # inside (button) can put text and tags (I | b | strong | img | br | etc) - that is not possible with a button created with the (input) element # NB: always specify the (type) attribute for a (button) element, - to tell browsers what type of button it is # can easily style buttons with CSS
05	select	None	Autofocus | disabled | Form Multiple | Name | required size Global | Event	# the (select) is used to create a drop-down list # the (select) is most often used in a form, to collect user input # the (name) attribute is needed to reference the form data after - the form is submitted (if omit (name), no data will be submitted)

				# (id) attribute is needed to associate the drop-down list with a label # (option) inside (select) define the available options in drop-down list # NB: always add the (label) tag for best accessibility practices
06	optgroup	None	Disabled | label Global | Event	# (optgroup) defines a group of related options in a (select), drop-down list # for long list of options, groups of related options are easy to handle
07	option	None	Disabled | label Selected | value Global | Event	# the (option) tag defines an option in a select/drop-down list # (option) elements go inside a (slelect | optgroup | datalist) # NB: (option) can be used without attribute usually need (value) attribute, - which indicates what is sent to the server on form submission # NB: for long list of options, can group related options within (optgroup)
08	label	Cursor: default	For | form Global | Event	# the (label) tag defines a label for an (input) element, like - - checkbox | color | date | datetime-local | email | file | mouth | url - number | password | radio | range | serach | tel | text | time | week - (meter | progress | select | textarea) element # proper use of labels with the elements above will benefir - screen reader users (rad out loud the label, and focused on element) - users who have difficulty clicking on very small regions (chekboxes) - because whe a user clicks the text within the (label), toggles input # NB: (for) attribute of (label) must be equal to (id) of related element - a (label) can also be bound to an element by placing it inside (lable)
09	fieldset	Display: block Margin-left: 2px Margin-right: 2px Padding-top: 0.35em Padding-bottom: 0.625em Padding-left: 0.75em Padding-right: 0.75em Border: 2px groove color	Disabled | form | name Global | Event	# (fieldset) tag is used to groups related elements in a form # (fieldset) tag draws a box around the related elements # NB: the (legend) is used to define a caption for the (fieldset)
10	legend	Display: block Padding-elft: 2px Padding-right: 2px Border: none	Global | Event	# the (legend) tag defines a caption for a (fieldset) element
11	datalist	Display: none	Global | Event	# (datalist) specifies a list of pre-defined options for (input) element # (datalist) is used to provide an autocomplete feature for (input) - users will see a drop-down list for pre-defined options as they input # the (id) must be equal to (input) tag's (list) attribute to binds them
12	output	Display: inline	For | form | name Global | Event	# the (output) tag is used to represent the result of a calculation - like one performed by a script

G11	Frames	Default CSS	Attributes	Definition
01	frame (not html5)	None	None	# defines a window (a frame) in a frameset - not in HTML5 # use the (iframe) tag to embed another document within the current HTML
02	frameset (not html5)	None	None	# defines a set of frames - not in HTML5 # use the (iframe) tag to embed another document within the current HTML
03	noframes (not html5)	None	None	# defines an alternate content that do not support frames - not in HTML5 # use the (iframe) tag to embed another document within the current HTML
04	iframe	Iframe: focus – Outline: none Iframe[seamless] – Display: block	Alow | allowfullscreen Allowpaymentrequest Height | loading | width Name | referrerpolicy Sandbox | src | srcdoc Globla | Event	# the (iframe) tag defines an inline frame # an inline frame is used to embed another document within the current HTML # NB: use CSS to style the (iframe)
05	embed	Embed: focus – Outline: none	Height | src | type Width Global | Event	# defines a container for an external (non-HTML) application # the (embed) tag defines a container for an external resource, - such as a web page, a picture, a media player, or a plug-in application # NB: most browsers no longer support Java Applets and plug-ins - activeX control are no longer supported in any browsers - support for shockwave flash has been turned off in modern browser # to display a picture, it is better to use the (img) tag - to display HTML, it is better to use the (iframe) tag - to display (video or audio), it is better to use the (video | audio) tags
06	object	Object: focus – Outline: none	Data | form | height Name | type | usemap Width | typemustmatch Global | Event	# (object) tag defines (embedded object) a container for external resource # external resource can be (web page | picture | media player | plug-in) # to embed a picture, it is better to use the (img) tag - to embed HTML, it is better to use the (iframe) tag - to embed (video | audio), it is better to use the (video | audio) tag # plug-ins - the (object) tag was originally designed to embed browser plug-ins - pug-ins are programs that extend standard functionality of browser - plug-ins have been used for many different purposes – - run java applets | run activeX controls | display flash movies - display maps | scan for viruses | verify a bank id # most browsers no longer support Java Applets and Plug-ins - activeX controls and shockwave flash are no longer supported in any browser
07	param	Display: none	Name | value Global | Event	# defines a parameter for an object # the (param) tag is used to define parameters for an (object) element

G12	Programming	Default CSS	Attributes	Definition
01	script	Display: none	Async | crossorigin Defer | integrity Nomodule Referrerpolicy Src | type Global	# defines a client-side script # the (script) tag is used to embed a client-side script (JS) # the (script) either contains scripting statements, - or it points to an external script file through the (src) attribute # common users for JS are image manipulation, form validation, and - Dynamic changes of content # NB: also look at the (noscript) for users that have disabled scripts in browser - or have a browser that doesn't support client-side scripting # in XHTML, content inside (script) is declared as (#PCDATA) instead (CDATA) - which means that entities will be parsed # this means that in XHTML, all special characters should be encoded, - or all content should be wrapped inside a CDATA section
02	noscript	None	Global	# defines an alternate content for users that do not support client-side scripts # the (noscript) tag defines an alternate content to be displayed to users - that have disabled scripts in their browser or have a browser that doesn't - support script # the (noscript) can be used in both (head | body) - when used inside (head), (noscript) could only contain (link | style | meta)
03	applet	None	None	# defines an embedded applet - not in HTML5 | use (embed | object) # plug-ins - plug-ins are programs that extend the standard functionality of browser - plug-ins have been used for many different purposes - run java applets | run activeX controls | display flash movies | display maps - scan for viruses | verify a bank id # most browsers no longer support Java Applets and Plug-ins - activeX controls are no longer supported in any browsers - the support for shockwave flash has also been turned off in modern browsers # if want to embed a video, use the (video | audio | embed | object) tag - To embed a picture, it is better to use the (img) tag - to embed a document, it is better to uset the (iframe)

01	Accept	Input	
02	Accept-charset	Form	
03	Action	Form	
04	Align	Not in HTML5	
05	Alt	Area | img | input	
06	Async	Script	
07	Autocomplete	Form | input	
08	Autofocus	Button | input | select | textarea	
09	Autoplay	Audio | video	
10	Bgcolor	Not in HTML5	
11	Border	Non in HTML5	
12	Charset	Meta | script	
13	Checked	Input	
14	Cite	Blockquote | del | ins | q	
15	Color	Not in HTML5	
16	Cols	Textarea	
17	Colspan	Td | th	
18	Content	Meta	
19	Controls	Audion | video	
20	Cords	Area	
21	Data	Object	
22	Datetime	Del | ins | time	
23	Default	Track	
24	Defer	Script	
25	Dirname	Input | textarea	
26	Disabled	Button | fieldset | input | optgroup Option | select | textarea	
27	Download	A | area	
28	Enctype	Form	
29	For	Label | output	
30	Form	Button | fieldset | input | label Meter | object | output | select Textarea	
31	Formaction	Button | input	
32	Headers	Td | th	
33	Height	Canvas | embed | iframe | img Input | object | video	
34	High	Meter	

35	Href	A | area | base | link	
36	Hreflang	A | area | link	
37	http-equiv	Meta	
38	Ismap	Img	
39	Kind	Track	
40	Label	Track | option | optgroup	
41	List	Input	
42	Loop	Audion | video	
43	Low	Meter	
44	Max	Input | meter | progress	
45	Maxlength	Input | textarea	
46	Media	A | area | link | source | style	
47	Method	Form	
48	Min	Input | meter	
49	Multiple	Input | select	
50	Muted	Video | audio	
51	Name	Button | fieldset | form | iframe Input | map | meta | object | output Param | select | textarea	
52	Novalidate	Form	
53	Open	Details	
54	Optimum	Meter	
55	Pattern	Input	
56	Placeholder	Input | textarea	
57	Popovertarget	Button | input	
58	Popvertargetaction	Buttion | input	
59	Poster	Video	
60	Preload	Audio | video	
61	Readonly	Input | textarea	
62	Rel	A | area | form | link	
63	Required	Input | select | textarea	
64	Reversed	Ol	
65	Rows	Textarea	
66	Rowspan	Td | th	
67	Sandbox	Iframe	
68	Scope	Th	
69	Selected	Option	
70	Shape	Area	

71	Size	Input | select	
72	Sizes	Img | link | source	
73	Span	Col | colgroup	
74	Src	Audio | embed | iframe | img | input Script | source | rtack | video	
75	Srcdoc	Iframe	
76	Srclang	Track	
77	Srcset	Img | source	
78	Start	Ol	
79	Step	Input	
80	Target	A | area | base | form	
81	Type	A | button | embed | input | Menu link | object | script | source | style	
82	Usemap	Img | object	
83	Value	Button | input | li | option | meter Progress | param	
84	Width	Canvas | embed | iframe | img Input | object | video	
85	Wrap	Textarea	

01	Accesskey	Specifies a shortcut key to activate/focus an element	
02	Class	Specifies one or more classnames for an element (refers to a class in a style-sheet)	
03	Contenteditable	Specifies whether the content of an element is editable or not	
04	Data-*	Used to store custom data private to the page or application	
05	Dir	Specifies the text direction for the content in an element	
06	Draggable	Specifies whether an element is draggable or not	
07	Enterkeyhint	Specifies the text of the enter-key on a virtual keyboard	
08	Hidden	Specifies that an element is not yet, or is no longer, relevant	
09	Id	Specifies a unique id for an element	
10	Inert	Specifies that the browser should ignore this section	
11	Inputmode	Specifies the mode of a virtual keyboard	
12	Lang	Specifies the language of the element’s content	
13	Popover	Specifies a popover element	
14	Spellcheck	Specifies whether the element is to have its spelling and grammar checked or not	
15	Style	Specifies an inline CSS style for an element	
16	Tabindex	Specifies the tabbing order of an element	

17	Title	Specifies extra information about an element	
18	Translate	Specifies whether the content of an element should be translated or not	

	Window		
01	Onafterprint	After the document is printed	
02	Onbeforeprint	Before the document is printed	
03	Onbeforeeunload	When the document is about to be unloaded	
04	Onerror	When an error occurs	
05	Onhaschange	When there has been changes to the anchor part of a URL	
06	Onload	After the page is finished loading	
07	Onmessage	When the message is triggered	
08	Onoffline	When the browser starts to work offline	
09	Ononline	When the browser starts to work online	
10	Onpagehide	When a user navigates away from a page	
11	Onpageshow	When a user navigates to a page	
12	Onpopstate	When the window's history changes	
13	Onresize	When the browser window is resized	
14	Onstorage	When a web storage area is updated	
15	Onunload	Once a page has unloaded (or the browser window has been closed)	
	Form		
16	Onblur	The moment that the element loses focus	
17	Onchange	The moment when the value of the element is changed	
18	Oncontextmenu	When a context menu is triggered	
19	Onfocus	The moment when the element gets focus	
20	Oninput	When an element gets user input	
21	Oninvalid	When an element is invalid	
22	Onreset	When the reset button in a form is clicked	
23	Onsearch	When the user writes something in a search field	
24	Onselect	After some text has been selected in an element	
25	Onsubmit	When a form is submitted	
	Keyboard		
26	Onkeydown	When a user is pressing a key	
27	Onkeypress	When a user presses a key	
28	Onkeyup	When a user releases a key	
	Mouse		
29	Onclick	On a mouse click on the element	

30	Ondblclick	On a mouse double-click on the element	
31	Onmousedown	When a mouse button is pressed down on a element	
32	Onmousemove	When a mouse pointer is moving while it is over an element	
33	Onmouseout	When a mouse pointer moves out of an element	
34	Onmouseover	When a mouse pointer moves over an element	
35	Onmouseup	When a mouse button is released over an element	
36	Onmousewheel	Deprecated – use the onwheel attribute instead	
37	Onwheel	When the mouse wheel rolls up or down over an element	
	Drag		
38	Ondrag	When an element is dragged	
39	Ondragend	At the end of a drag operation	
40	Ondragenter	When an element has been dragged to a valid drop target	
41	Ondragleave	When an element leaves a valid drop target	
42	Ondragover	When an element is being dragged over a valid drop target	
43	Ondragstart	At the start of a drag operation	
44	Ondrop	When dragged element is being dropped	
45	Onscroll	When an element’s scrollbar is being scrolled	
	Clipboard		
46	Oncopy	When the user copies the content of an element	
47	Oncut	When the user cuts the content of an element	
48	Onpast	When the user pastes some content in an element	
	Media		
49	Onabort	Run on abort	
50	Oncanplay	When a file is ready to start playing (buffered enough to being)	
51	Oncanplaythrough	When a file can be played all the way to the end without pausing for buffering	
52	Oncuechange	When the cue changes in a (track) element	
53	Ondurationchange	When the length of the media changes	
54	Onemptied	When something bad happens and the file is suddenly unavailable (unexpectedly disconnects)	
55	Onended	When the media has reach the end (a useful event for messages)	
56	Onerror	When an error occurs when the file is being loaded	
57	Onloadeddata	When media data is loaded	
58	Onloadedmetadata	When meta data (dimensions and duration) are loaded	
59	Onloadstart	Just as the file begins to load before anything is actually loaded	
60	Onpause	When the media is paused either by the user or programmatically	
61	Onplay	When the media is ready to start playing	
62	Onplaying	When the media actually has started playing	
63	Onprogress	When the browser is in the process of getting the media data	
64	Onratechange	Each time the playback rate changes (switches ot a slow motion or fast forward mode)	

65	Onseeked	When the seeking attribute is set to false indicating that seeking has ended	
66	Onseeking	When the seeking attribute is set to true indicating that seeking is active	
67	Onstalled	When the browser is unable to fetch the media data for whatever reason	
68	Onsuspend	When fetching the media data is stopped before it is completely loaded for whatever reason	
69	Ontimeupdate	When the playing position has changed (fast forwards to a different point in the media)	
70	Onvolumechange	Each time the volume is changed	
71	Onwaiting	When the media has paused but is expected to resume	
	Misc		
72	Ontoggle	When the user opens or close the (details) element	